Section 6.2 revision: results summary

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Proposed Section 6.2: margin adjustment against a frozen ECDF

Three data generating processes from Section 6.1 (VARMA, VARCH, VARMA Copula), sample sizes $T = 25, 50, 100, 200$, 100 replicates each with parameters redrawn every replicate. Both arms are forecast from the *same* posterior fit with the RNG state restored between calls, so the latent forward simulations are identical and the only difference is how the margin is treated.

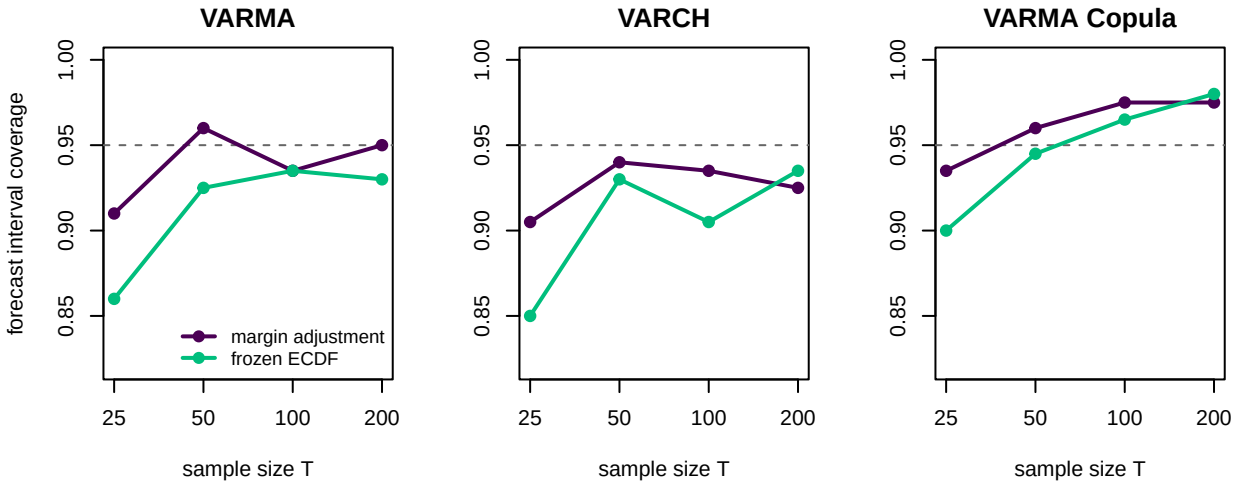


Figure 1: Coverage of the 95% forecast interval at $h = 1$, against training sample size. Each point is 200 forecasts (100 replicates, two variables). Dashed line is nominal.

Table 1: Forecast interval coverage at $h = 1$, pooled across the three DGPs and both variables (600 forecasts per cell). Nominal is 0.95.

T	frozen ECDF	margin adj.	ECDF shortfall	adj. shortfall	McNemar p
25	0.870	0.917	0.080	0.033	0.000
50	0.933	0.953	0.017	-0.003	0.008
100	0.935	0.948	0.015	0.002	0.021
200	0.948	0.950	0.002	0.000	1.000

The benefit is concentrated in small samples. At $T = 25$ the frozen margin undercovers by 8 points and the estimated margin by 3.3, so estimating F closes about three fifths of the gap. By $T = 50$ the estimated margin is at nominal while the frozen one remains about 1.5 points under, a gap that an exact McNemar test on the paired indicators still detects at $T = 100$, and by $T = 200$ the two are indistinguishable.

This is consistent with the real-data applications rather than in tension with them. CRPS is a wash in both. The macro vintages run $T = 84$ to 199 and the crime series is longer still, so both sit at the upper end of the range in which the margin adjustment changes anything. The coverage effect in the applications is larger than this simulation produces at matched sample size, which is what one would expect given that none of these three DGPs is as persistent as the macro level series.

Point accuracy and CRPS are unaffected; the gain is specific to interval calibration. Supporting detail, together with a separate finding on the calibration of the margin posterior itself, is in the appendix.

Appendix: supporting detail

A.1 is the one item that might warrant promotion to the main text.

A.1 Calibration of the margin posterior

The margin adjustment's own 95% credible band for F (the object Figure 4 in Section 6.1 plots) covers the truth less and less often as the sample grows.

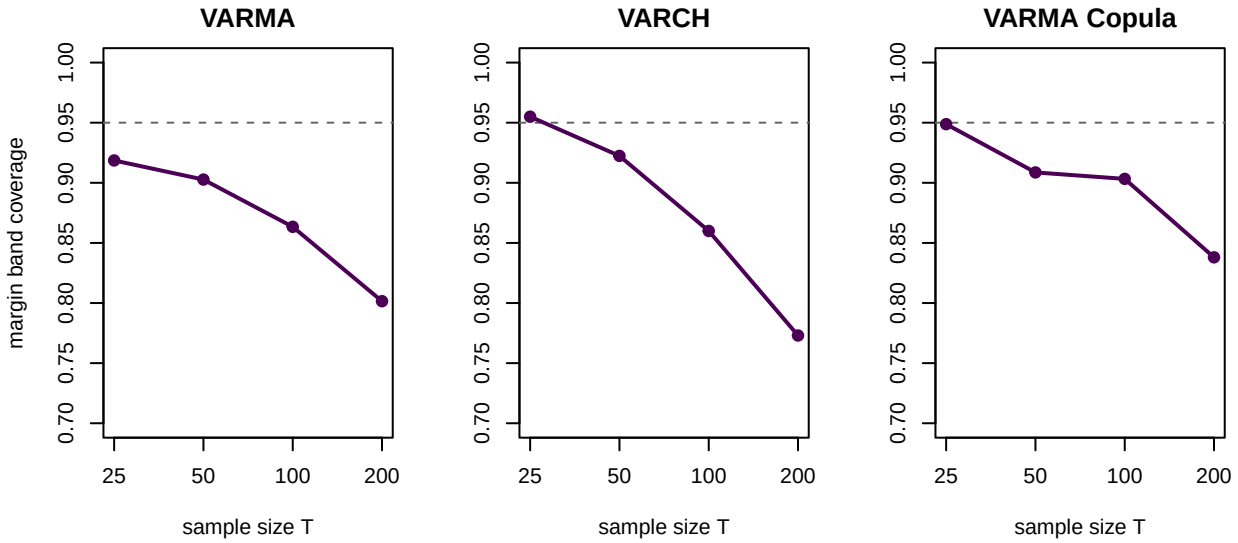


Figure 2: Coverage of the margin adjustment's own 95% pointwise credible band for the true F. The frozen ECDF has no band. Dashed line is nominal.

Table 2: Coverage of the margin adjustment's 95% pointwise credible band for the true F.

T	VARCH	VARMA	VARMA Copula
25	0.955	0.919	0.949
50	0.922	0.903	0.909
100	0.860	0.863	0.903
200	0.773	0.802	0.838

The decline is monotone in all three DGPs and is not a sampling artifact. A dedicated run at four times the chain length returns coverage still far from nominal.

Table 3: Chain-length check at $T = 200$. Absolute L1 levels are not comparable across the two runs because they draw different DGP parameters; the within-run comparison and the band coverage are.

DGP	chain	ECDF L1	adj. L1	band coverage	reps
VARCH	10,000 sweeps	0.02293	0.02568	0.773	100
VARCH	40,000 sweeps	0.02094	0.02344	0.856	20
VARMA	10,000 sweeps	0.03004	0.03178	0.802	100
VARMA	40,000 sweeps	0.03697	0.03647	0.780	20

Two claims should be separated here. The posterior does concentrate around the truth as T grows, which is what Theorem 3 and the text of Section 6.1 assert, and the table in A.2 shows that concentration directly. What fails is calibration of the band around that concentration. That is the expected behaviour of a pseudo-posterior: the rank likelihood is not a likelihood, so its curvature has no reason to match sampling variability, and the mismatch becomes visible precisely as the posterior tightens.

Note also that the predictive intervals remain calibrated, because the predictive integrates over the margin posterior and the latent dynamics together.

A.2 Point accuracy and density scores

Table 4: Mean absolute error of the estimated margin against the true F. Both arms converge. A positive difference means the margin adjustment is the less accurate point estimate.

DGP	T	frozen ECDF	margin adj.	diff %
VARCH	25	0.0641	0.0633	-1.4
VARCH	50	0.0439	0.0457	4.1
VARCH	100	0.0315	0.0342	8.7
VARCH	200	0.0229	0.0257	12.0
VARMA	25	0.0788	0.0742	-5.8
VARMA	50	0.0527	0.0561	6.6
VARMA	100	0.0376	0.0399	6.0
VARMA	200	0.0300	0.0318	5.8
VARMA Copula	25	0.0615	0.0636	3.5
VARMA Copula	50	0.0423	0.0470	11.1
VARMA Copula	100	0.0288	0.0305	5.9
VARMA Copula	200	0.0203	0.0231	14.0

Both arms converge at the expected rate, which is the direct check that nothing in the sampler is broken. The margin adjustment is a few percent less accurate as a point estimate from $T = 50$ onward, though that gap did not reproduce on one DGP in the chain-length check and should not be leaned on.

Table 5: CRPS at $h = 1$, paired difference. Positive means freezing F is worse.

DGP	T	mean %	s.e.
VARCH	25	-2.6	0.8
VARCH	50	-1.6	0.6
VARCH	100	-0.7	0.4
VARCH	200	-1.5	0.4
VARMA	25	+6.3	2.8
VARMA	50	+3.6	1.9
VARMA	100	+4.5	2.0
VARMA	200	+5.5	2.8
VARMA Copula	25	+0.0	1.8
VARMA Copula	50	+7.5	6.8
VARMA Copula	100	-0.1	0.9
VARMA Copula	200	+0.0	0.9

CRPS is a wash with no pattern in sample size, matching the real data where the median change across both applications was under one percent. The benefit of estimating the margin is in calibration, not sharpness or the density score.

A.3 Mechanism

An empirical CDF can place no probability outside the range it has already seen, so the penalty for freezing F should scale with how often the realised value falls outside the training range.

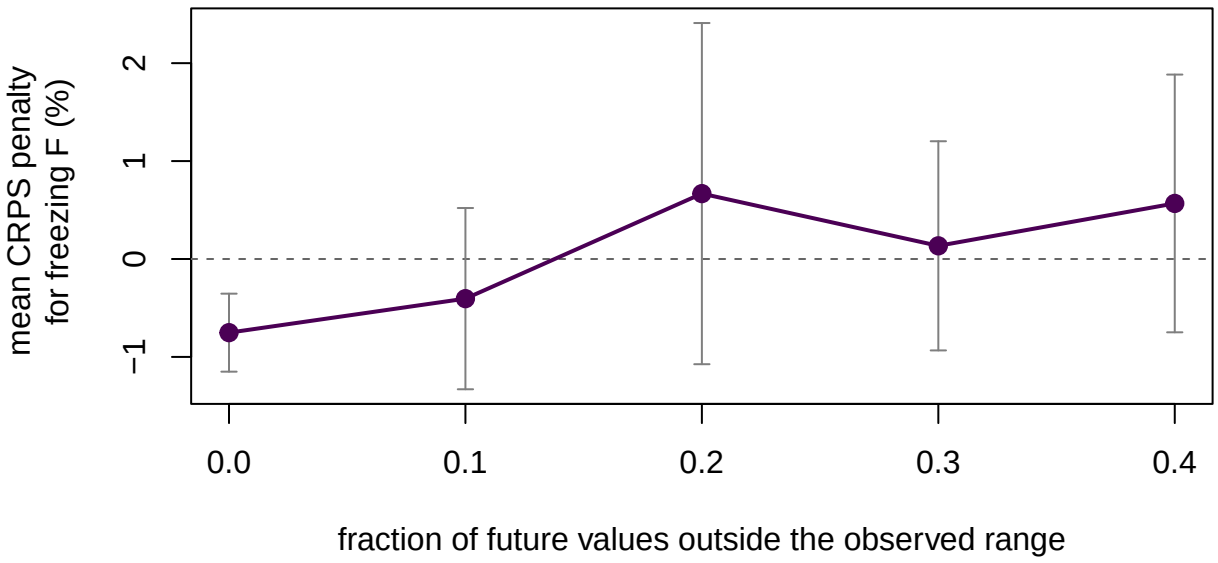


Figure 3: Mean CRPS penalty for freezing F at each observed out-of-range rate, with two standard errors. Rates represented by fewer than ten cells are dropped.

Correlation between the two: **0.238**. Only the point at an out-of-range rate of zero excludes no change, and it falls slightly below it: when nothing leaves the observed range, freezing F is marginally the better estimator.

Rates above 0.4 appear in too few cells to plot, because none of these three DGPs produces a series persistent enough to leave its own support often. That is the same limitation behind the observation in the main text that the coverage effect in the applications exceeds what this simulation produces at matched sample size. The macro level series `ur`, `ffr` and `capu` are far more persistent than anything generated here. Testing the mechanism would require a DGP with a root near unity, which is a change to the design rather than to the analysis.

A.4 Design

Differences from the Section 6.2 this replaces, all of which affect whether the numbers can be relied on:

- **100 replicates** per cell with DGP parameters redrawn each time, against a single realisation. The old script declared `nrep = 10` and never used it.
- `burn = 5000, thin = 5`, matching the consistency study in 6.1, against `burn = 0`.
- **Paired arms.** Both forecasts come from one posterior fit with the RNG state restored between calls, so the latent forward draws are identical and only the margin treatment differs.
- **Burn-in discarded from the simulators.** `simulate_stationary_varma` draws its initial lags iid from the stationary marginal, which gets the marginals right but not the autocorrelation between them, and the transient decays at the companion spectral radius. At $T = 25$ that would have contaminated the cell the section exists to measure.
- **Per-variable spline control.** The VARMA Copula carries a Poisson variable and a skew-t variable in one series; the step inversion is used for the former and the monotone spline for the latter.
- **Margin accuracy scored against the true** F, which the closed-form stationary distributions make possible and which the old section never did.